AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM358
Machine Learning & Artificial Intelligence

Final Examination
Spring Semester 2025/2026

Date: 9/ June/ 2026

Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed:** 3 hours

Instructions:

- Total Marks: **100**
- This exam consists of 2 parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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PART 1

Multiple Choice Questions **(20 marks)**

Answer the following MCQ by selecting **the best** answer for each question. Write your **answer letter** in the answer sheet [2 marks for each point].

1 ..... can be defined as "The sifting out of units whose characteristics and behavior make them less capable of responding successfully to the problems thrown up by their environment"

A adaptation B selection
C interaction D emergence

2 ..... aim to find good solutions within a reasonable length of computing time.

A exact B optimization
C combinational optimization D approximate algorithms

3 A ..... is a way of identifying some part of a genome, a template.

A chromosome B schema
C protein D genome

4 ..... is to use an object characteristics to identify which class/category it belongs to.

A training B Clustering
C Classification D None of the above

5 In ……. architecture there are n layers, with all units in a layer connected to all units in the next layer, but not to units in other layers or in the same layer. Activity feeds forward from the first layer through to the final layer.

A Lattice B Layered
C Recurrent D None of the above

6 insect movement is an example of .....

A interaction with other agents B internal interaction
C interaction with the environment D None of the above

7 ..... can be defined as "The ability of some unit to alter its longer-term behavior patterns in response to challenges from its environment"

A emergence B selection
C interaction D adaptation

8 ..... involve building a possible solution step by step, at each step picking the most possible candidate from the set of possibilities

A adaptation B Construction algorithms
C Local search algorithms D swarm intelligence

9 ..... can be defined as "is the collective behavior of decentralized, self-organized systems, natural or artificial"

A swarm intelligence B selection
C adaptation D interaction

10 ..... is the number of fixed positions in that schema.

A defining length B order
C dimension D None of the above

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PART 2

This part contains 4 questions with a total of 80 marks.
Answer all questions

Short problems/questions

Question 2, [20 marks]

A. What are the key differences between LSTMs and standard RNNs? **[5 marks]**

B. **[15 marks]**

Given an RNN with series length 2. Assume the following weights:

W_{xh}=

-0.27	-0.88	1.46
-0.94	0.60	-1.11
1.88	-0.13	-1.05
0.28	0.33	0.98
0.35	0.58	0.58

W_{hh}=

-0.44	0.66	0.09	-0.91	0.98
-0.29	-1.77	-1.40	0.34	2.10
0.66	-0.22	-0.16	1.66	0.81
-0.14	-0.30	1.55	0.30	1.07
1.66	1.08	-0.17	-1.24	-0.57

W_{bh}=

-0.96
0.00
-0.68
-0.73
0.66

W_{by}=

1.05
0.74
1.43

W_{hy}=

-0.59	-0.65	1.10	-1.61	0.22
-0.93	0.83	0.31	-1.00	0.02
-0.52	-0.11	-0.00	0.92	0.30

Assume the following input series are presented to the network in sequence:

X1	X2
0.00	1.00
1.00	0.00
0.00	0.00

Assume Hyperbolic activation for the hidden layer and Linear activation for the output layer

I. Calculate the hidden state value of the network at every input.

II. Calculate the final output at the end of the series.

Question 3, [20 marks]

Given the following solutions from GA population with their fitness:

A 4.000000
B 21.000000
C 28.000000
D 14.000000
E 3.000000

a. Calculate the probability of selecting each solution
b. Draw the roulette wheel representing the solution set
c. Assume the following stream of random numbers:[0.061614,0.776188,0.002714,0.606765,0.356901,], what will be the solutions entering the mating pool?

b. Having the following two chromosomes: [5 marks]

011110010
111100100

Apply crossover at point 3.

c. Given genome A:00101111001 [5 marks]

Write the output of the genome A after applying the mutation operator. Assume a mutation probability Pm= 0.16 and given the following sequence of random numbers corresponding to

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each chromosome:0.358952, 0.028651, 0.767375, 0.237384, 0.665913, 0.748491, 0.728439, 0.814288, 0.740083, 0.644517, 0.099002,

Question 4, [20 marks]

A. [10 marks]

Given the weight matrix for layer 1 in a neural net as:

0.23	0.15	0.31	0.27
0.71	0.02	0.16	0.90
0.43	0.42	0.99	0.27
0.42	0.95	0.85	0.85
0.93	0.73	0.67	0.42

The weight matrix for output layer:

0.09	0.64	0.28	0.17	0.55
0.94	0.99	0.83	0.18	0.65
0.82	0.63	0.93	0.95	0.99
0.93	0.43	0.95	0.63	0.81

and given training points:

3
2
1
1

with its corresponding desired output:

0.3
0.1
0.8
0.6

Assume that the actual outputs on nodes 1,2,3,4,5 in the hidden layer are: 0.83,0.96,0.97,0.99,1.00 in order.

Assume that ei on the output nodes 1,2,3,4 in the output layer are: -0.07,-0.03,-0.00,-0.01 in order.

Assume that ej on the nodes 1,2,3,5 in the hidden layer are: -0.01,-0.00,-0.00,-0.00 in order.

and assuming a learning rate of 0.8. Using backpropagation algorithm to update the neural weights for 1 iteration. Assume an activation function of type 1/(1+exp(-net)) for both layers

What is the weight change for weights attached to node 2 in the output layer= .....

a. What is the updated weights for weights attached to node 2 in the output layer= .....
b. What is ej on node 4 in the hidden layer= .....
c. What is the weight change for node 1 in the hidden layer= .....
d. What is the updated weight for node 1 in the hidden layer layer= .....

B. Given the following schema: **[7.5 marks]**

0100*001*10

I. Find the solutions matching this schema

II. The schema defining length

III. The schema order

Question 5, [20 marks]

A. [15 marks]

SAMPLE

Given the following particle positions in a PSO minimization optimization process at time t of the optimization:

X_1	X_2	X_3	X_4
-1	0	-1	1
1	-1	-1	1
2	1	2	-1
1	0	1	1

and given that each particle's attached velocity is:

V_1	V_2	V_3	V_4
-0.6	-0.7	-0.3	0.1
-0.1	-0.2	1.4	0.2
0.5	-1.7	0.1	-0.6
0.6	-0.1	0.5	0.5

and given each particle's local best as:

P_1	P_2	P_3	P_4
1	-2	-1	1
0	1	1	-1
1	-1	-2	-1
1	1	0	-1

assuming the global best is:[-1,1,-2,0,] Complete the following missing items that are calculated at time t+1 of the optimization. Assume $\Phi_1=0.3$ and $\Phi_2=0.5$ The function to be minimized is:$3x_1^1+6x_2^4+8x_3^5+9x_4^6$

V_1=

0.000000 -0.400000 -1.800000 0.100000

V_2=.....

V_3=

-0.300000 3.000000 -3.100000 -0.300000

V_4=

-0.900000 -0.400000 -1.100000 -0.600000

X_1=

-1.000000 0.600000 0.200000 1.100000

X_2=

-1.800000 0.400000 -2.800000 0.200000

X_3=.....

X_4=

0.100000 0.600000 -2.100000 0.400000

P_1=.....

P_2=

-1.800000 0.400000 -2.800000 0.200000

P_3=

-1.000000 1.000000 -2.000000 0.000000

P_4=

0.100000 0.600000 -2.100000 0.400000

P_g=.....

B. Explain the impact of different initial weights and learning constants on the training process of an artificial neural network. How does the order of presentation of patterns influence the learning outcome? [5 marks]

End of questions

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